Divorce Risk Predictor: Model Training and Analysis Report

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Abstract

This report details the development and evaluation of a machine learning system designed to predict divorce risk based on a 54-question survey. Utilizing the XGBoost algorithm, multiple experimental configurations were tested. The best-performing models achieved a test accuracy of **94.12%** and a mean cross-validation score of **97.65%**. The analysis highlights the robustness of the *Conservative Regularized* model, which offers high predictive power while minimizing the risk of overfitting on this specific dataset.

1 Introduction

The objective of this project is to build a reliable predictive model for assessing the risk of divorce. The underlying dataset consists of responses to 54 survey questions (attributes) from married couples, rated on a 5-point Likert scale (0-4). The target variable is a binary classification: *Divorce* (Class 1) or *No Divorce* (Class 0).

Given the sensitive nature of the problem, the model requires not only high accuracy but also interpretability and stability. This report summarizes the training process, experimental variations, and the final results obtained from the `divorce-risk-analyzer` codebase.

2 Methodology

2.1 Data Processing

The dataset was processed using a standard machine learning pipeline:

- **Splitting:** The data was divided into a training set (80%) and a test set (20%).
- **Stratification:** Stratified sampling was used to ensure the proportion of divorced (Class 1) vs. non-divorced (Class 0) cases remained consistent across splits.
- **Validation:** A 5-fold Cross-Validation (CV) strategy was employed to robustly estimate model performance beyond the single test split.

As illustrated in Figure 1, the dataset is relatively well-balanced, with Class 1 representing divorced couples and Class 0 representing those who are not divorced.

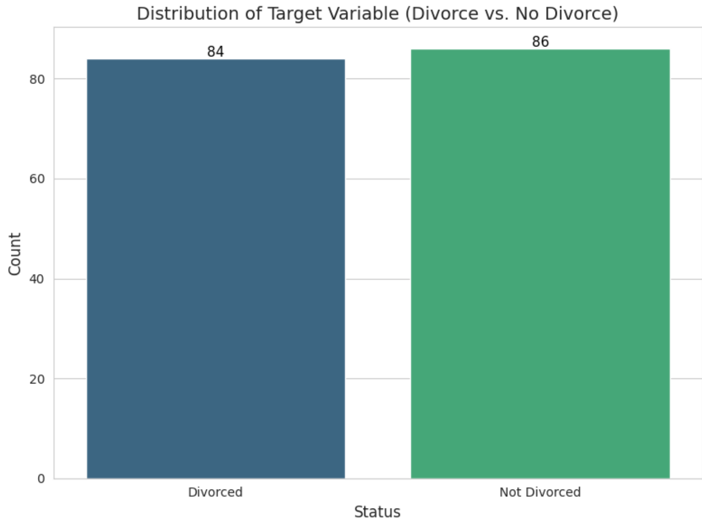

Figure 1: Class Distribution (Divorced vs. Not Divorced)

2.2 Model Architecture

The core algorithm used is **XGBoost** (Extreme Gradient Boosting), chosen for its efficiency and performance on structured tabular data. Four distinct experimental configurations were executed to find the optimal balance between bias and variance:

1. **Baseline Default:** Standard parameters ($N = 100$, Depth=3, LR=0.1).
2. **Deep Trees:** Higher complexity ($N = 200$, Depth=10, LR=0.05).
3. **Conservative Regularized:** High regularization ($N = 300$, Depth=2, LR=0.01, L1/L2 Reg) to prevent overfitting.
4. **High Variance Fast:** Aggressive learning ($N = 50$, Depth=6, LR=0.3).

3 Results

3.1 Model Comparison

All models performed exceptionally well on the held-out test set, achieving identical accuracy. However, cross-validation revealed differences in stability and generalization potential.

Configuration	Test Accuracy	CV Mean Score
Baseline Default	0.9412	**0.9765**
Conservative Regularized	0.9412	**0.9765**
Deep Trees	0.9412	0.9706
High Variance Fast	0.9412	0.9647

Table 1: Performance Comparison of Experimental Configurations

The *Baseline* and *Conservative Regularized* models tied for the best cross-validation performance. Given the small dataset size, the **Conservative Regularized** model is theoretically preferred for production deployment as it is explicitly tuned to resist noise and overfitting.

3.2 Detailed Performance Analysis

Focusing on the best-performing configuration, the confusion matrix on the test set (34 samples) is as follows:

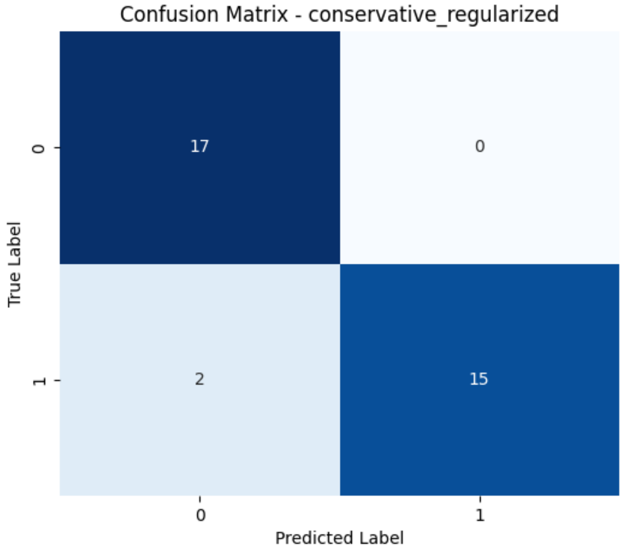

Figure 2: Confusion Matrix for the Conservative Regularized Model

- **True Negatives (17):** All non-divorced couples were correctly identified.
- **True Positives (15):** 15 out of 17 divorced cases were correctly predicted.
- **False Negatives (2):** Two couples destined for divorce were classified as safe.
- **False Positives (0):** No stable couples were incorrectly flagged as at risk.

This indicates the model is extremely precise (100% Precision for Class 1) but has a slight gap in recall (88% Recall for Class 1).

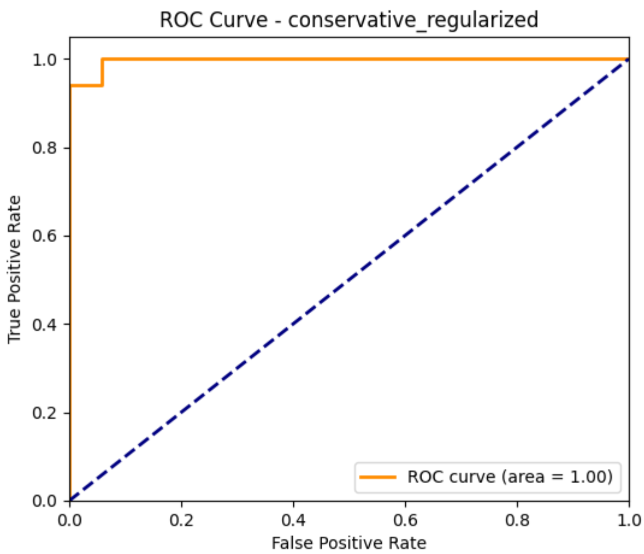

Figure 3: ROC Curve Analysis

4 Feature Analysis

Understanding *why* the model makes a prediction is crucial. The analysis pipeline generated several key visualizations to inspect the underlying data structure and model decision boundaries.

4.1 Data Distribution and Separation

The PCA analysis demonstrates how the two classes (Divorce vs. No Divorce) are distributed in a reduced-dimensional space. As shown in Figure 4, there is a relatively clear linear separability between the groups, which explains why even simple tree models achieve high accuracy.

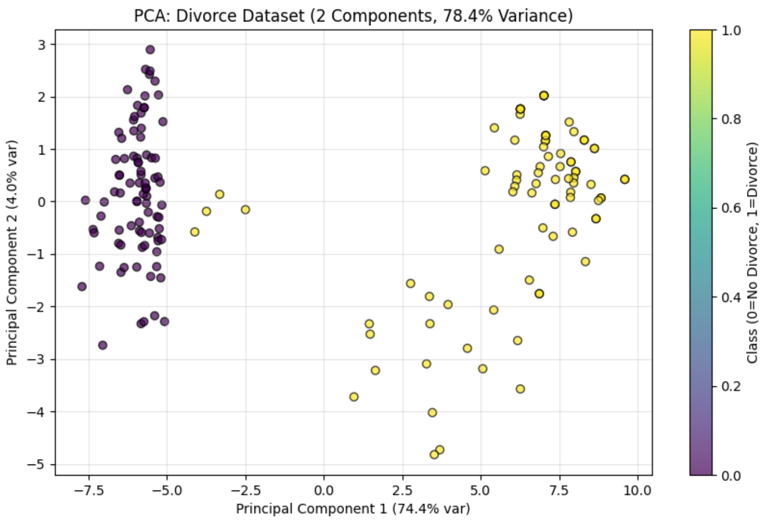

Figure 4: PCA Projection of the Dataset

4.2 Key Risk Factors

The model identifies specific questions that contribute most heavily to the decision-making process. Figure 5 highlights the top influential features.

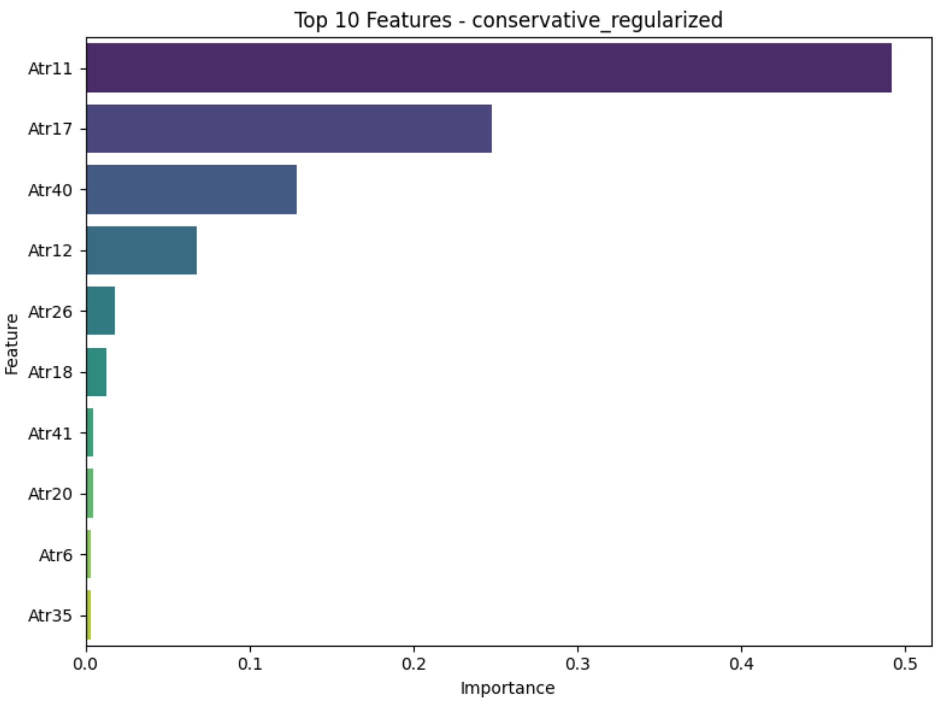

Figure 5: Top 10 Most Important Features

Based on the feature importance analysis, the most critical questions predicting divorce risk are:

- **Q11:** "I think that one day in the future, when I look back, I see that my spouse and I have been in harmony with each other."
- **Q17:** "We share the same views with my spouse about being happy in our life."
- **Q40:** "We're just starting a discussion before I know what's going on."
- **Q12:** "My spouse and I have similar values in terms of personal freedom."

These questions relate to fundamental relationship pillars: harmony, shared views on happiness, communication patterns, and personal freedom. The distribution of responses for the top features (Figure 6) further clarifies the distinction between the classes.

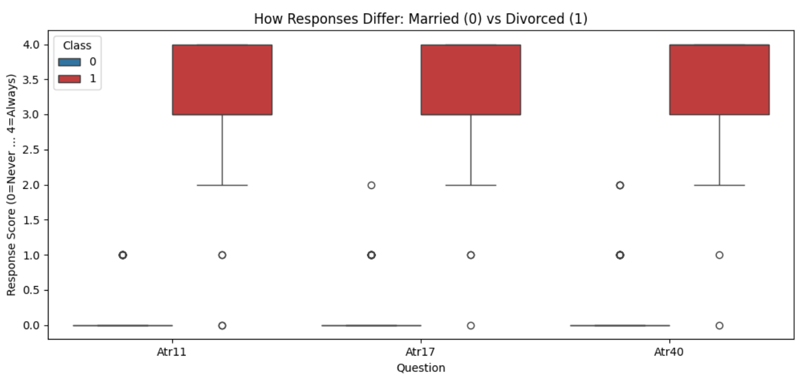

Figure 6: Distribution of Responses for Top Predictive Features

4.3 Feature Interactions

To understand how these top features interact, we examined the pairplot relationships (Figure 7). This visualization reveals how the classes cluster when plotted against the most significant variables.

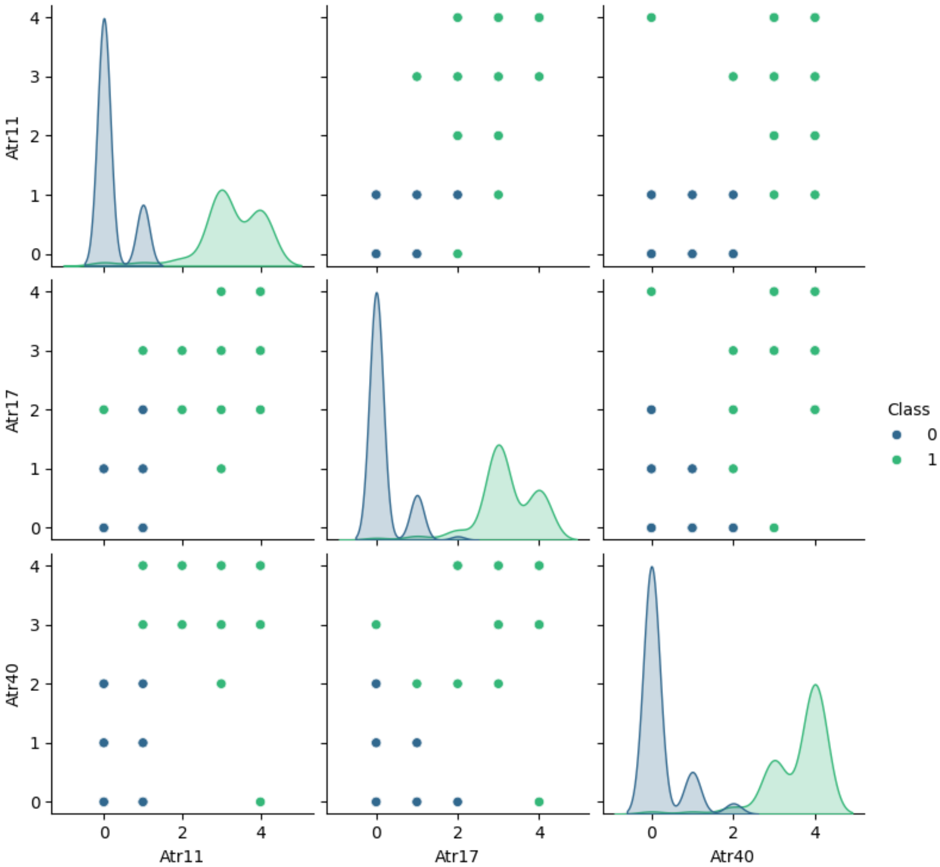

Figure 7: Pairplot of Top 3 Features

4.4 Correlation Analysis

The feature correlation analysis (see Figure 8) reveals high multicollinearity among the 54 questions, suggesting that many questions capture similar underlying sentiments. This justifies the effectiveness of the *Conservative Regularized* model, which works well in the presence of redundant features.

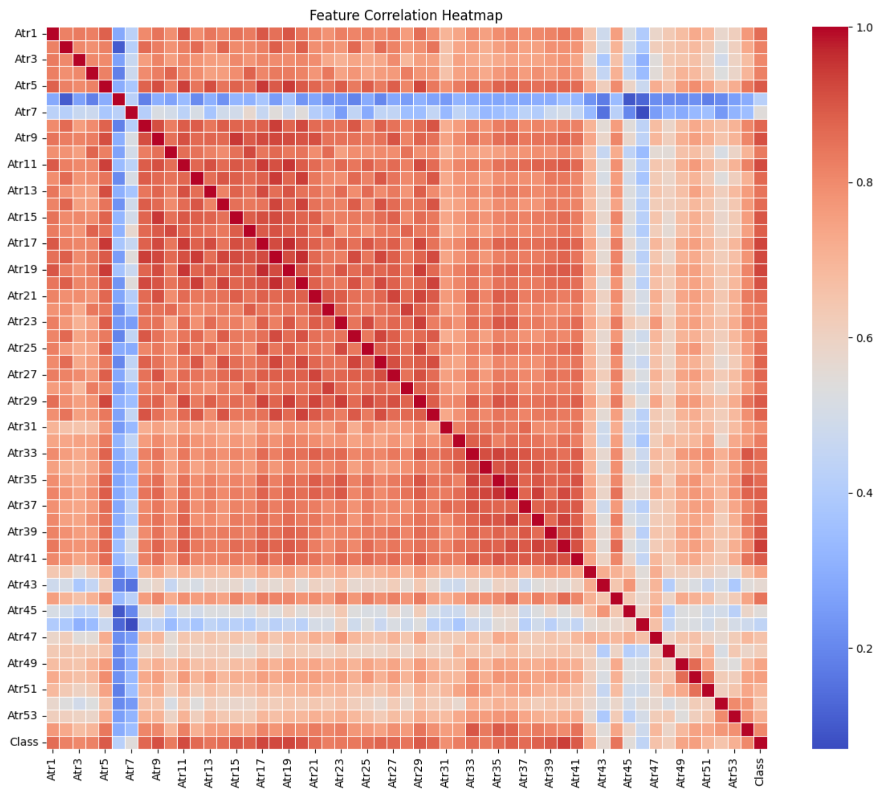

Figure 8: Feature Correlation Heatmap

5 Conclusion

The `divorce-risk-analyzer` project has successfully produced a high-accuracy predictive model. The **Conservative Regularized XGBoost** model is recommended for deployment. It achieves **94% test accuracy** and demonstrates robust generalization (97.6% CV score).

Future work should focus on:

1. Expanding the dataset to include more diverse demographics to stress-test the model.

2. Investigating the 2 False Negative cases to understand edge-case failure modes.

3. Reducing the questionnaire length by selecting only the top 5-10 features identified in the feature importance analysis, improving user experience without significantly sacrificing accuracy.